Kernix 0.13.0 / Performance brief

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

51 workloads

179 observations

Source precision retained

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

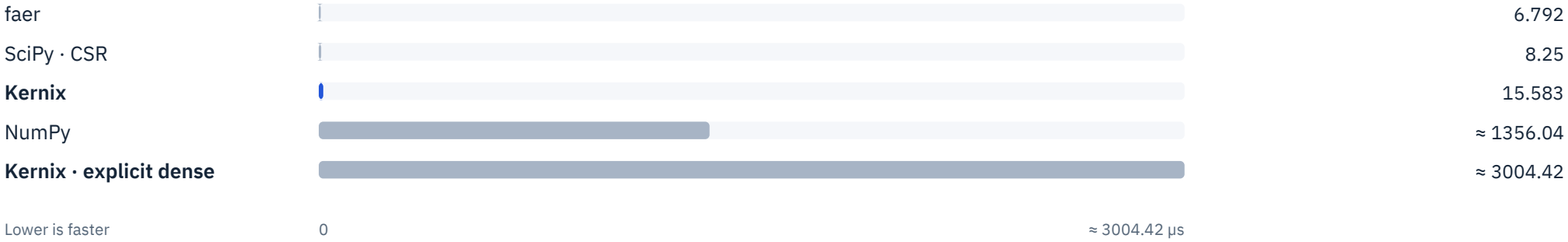
Kernix 0.13.0 / Sparse matrices & memory

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Sparse matrix × vector

Per-call median

2048 × 2048 · five diagonals · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Sparse × dense matrix

Per-call median

512 × 512 · 32 RHS · five diagonals · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Kernix 0.13.0 / Sparse matrices & memory

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Sparse × dense matrix

Per-call median

512 × 512 · 32 RHS · five diagonals · f32 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Sparse × sparse matrix

Per-call median

256 × 256 · five diagonals each · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Kernix 0.13.0 / Sparse matrices & memory

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Automatic CSR construction

Per-call median

512 × 512 · clone + scan + compress · f64 · Construction · dense copy or compression · preparation costs included



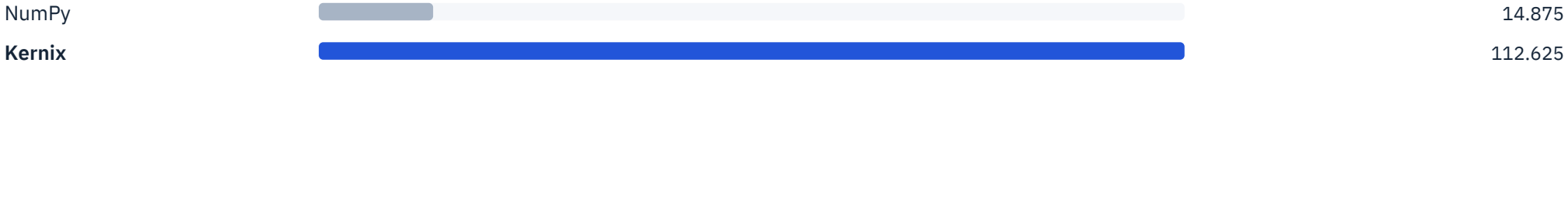
Kernix 0.13.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication · transposed view

Per-call median

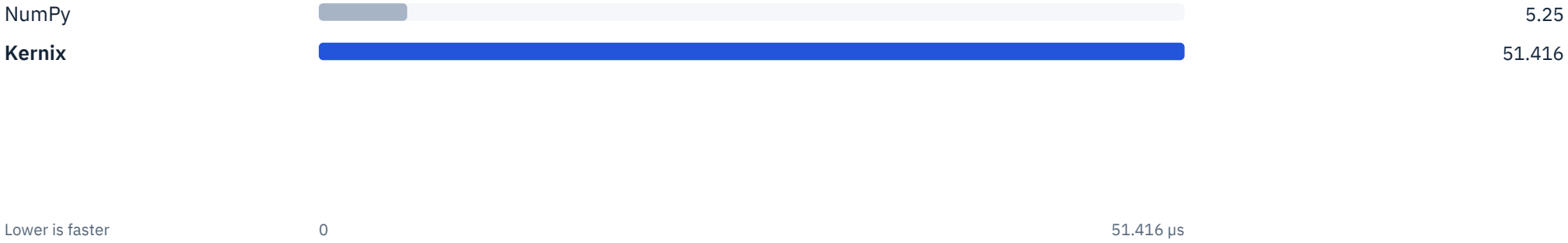
128 × 128 · 128 × 128 · f64 · 1 worker · borrowed transpose · allocated output · independent operands



Matrix multiplication · transposed view

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · borrowed transpose · allocated output · independent operands



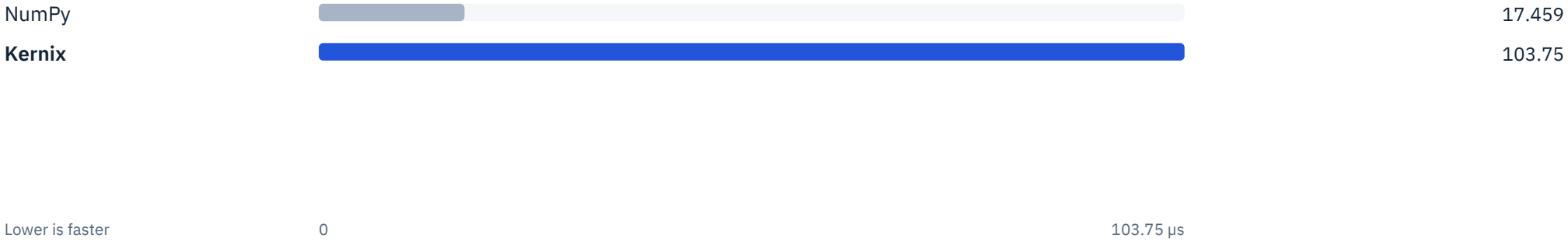
Kernix 0.13.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Independent matrix operands

Per-call median

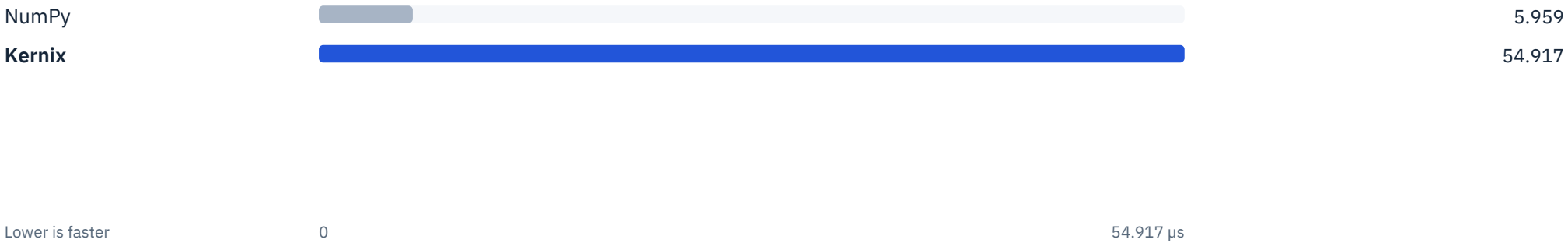
128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



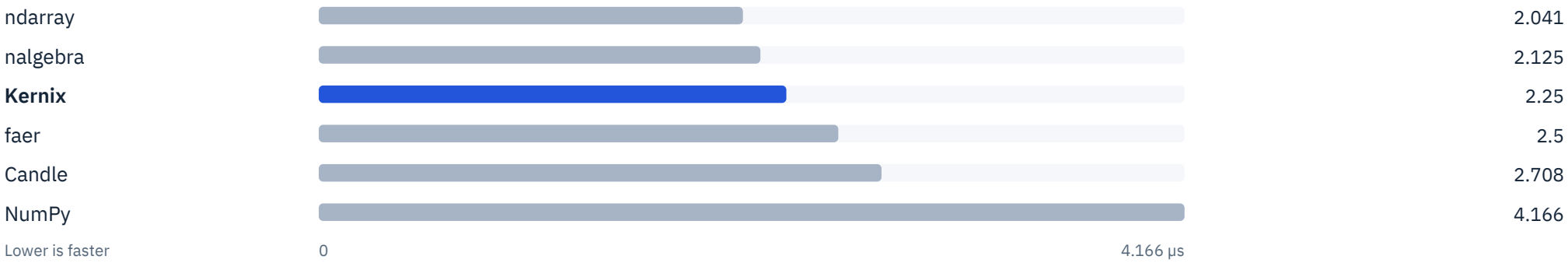
Kernix 0.13.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



Kernix 0.13.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



Kernix 0.13.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



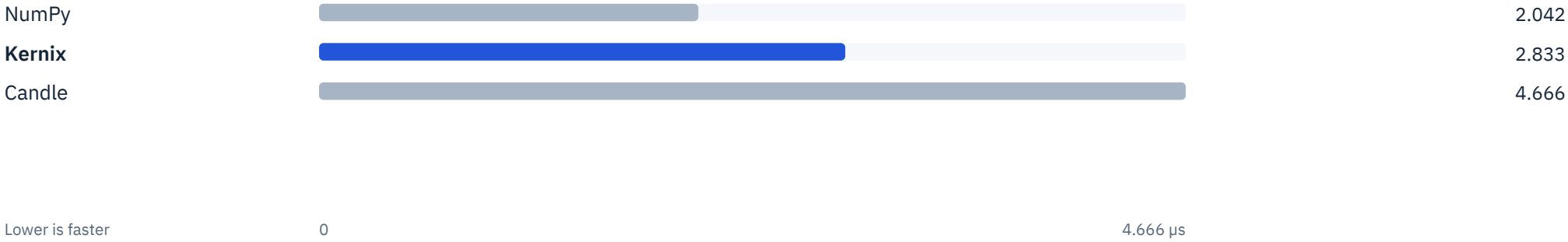
Kernix 0.13.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Batched multiplication

Per-call median

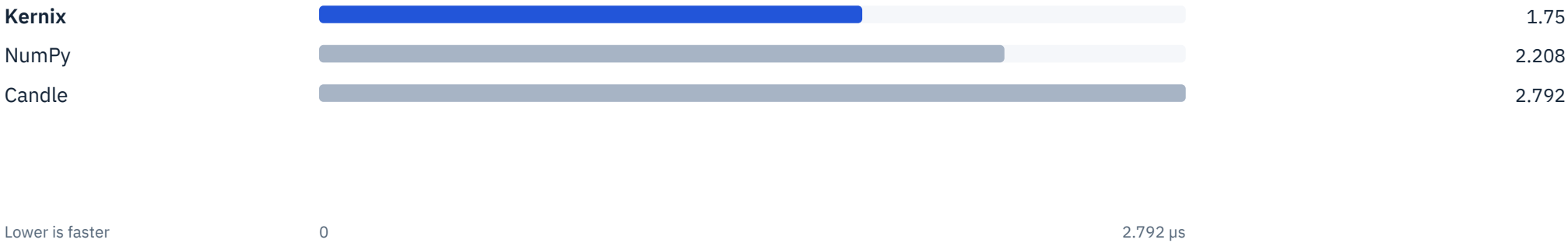
4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



Kernix 0.13.0 / Explicit parallel execution

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.13.0 / Explicit parallel execution

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



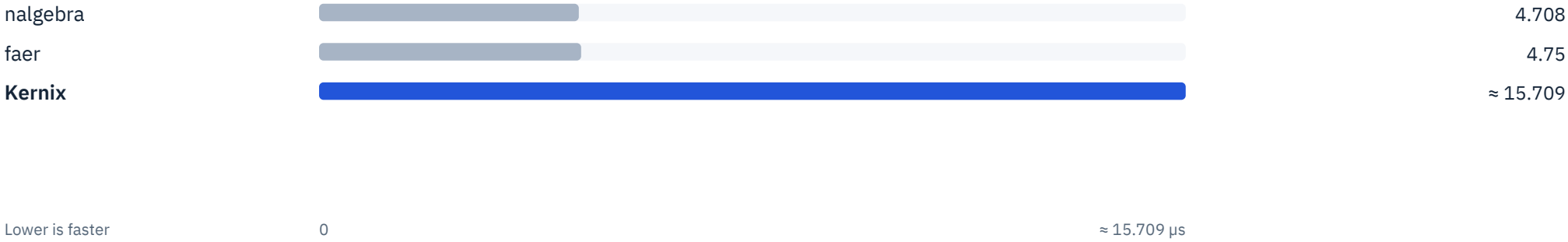
Kernix 0.13.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky solve · reused factor

Per-call median

128 × 128, 1 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Cholesky solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



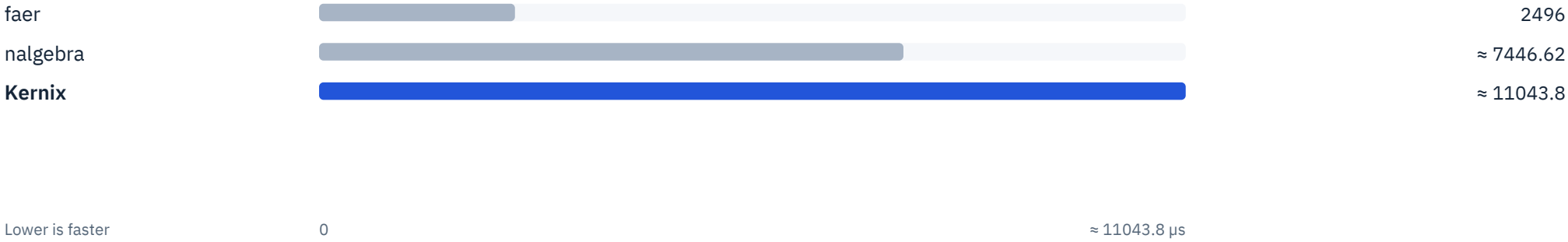
Kernix 0.13.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky solve · reused factor

Per-call median

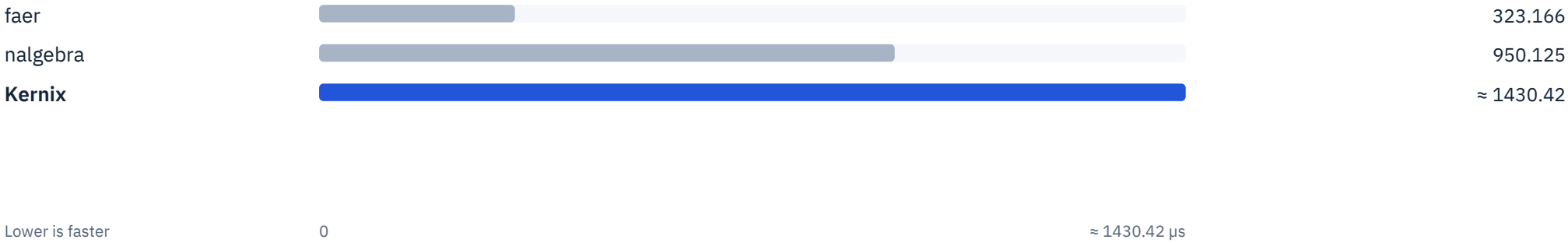
512 × 512, 128 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



LU solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



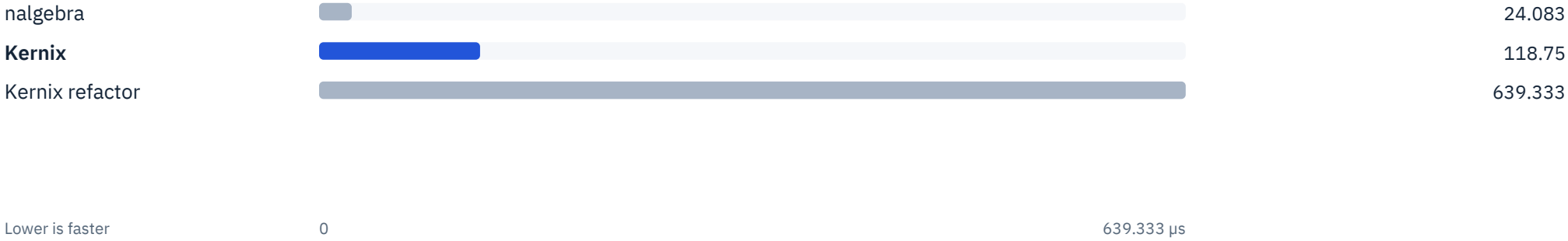
Kernix 0.13.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky rank-one update

Per-call median

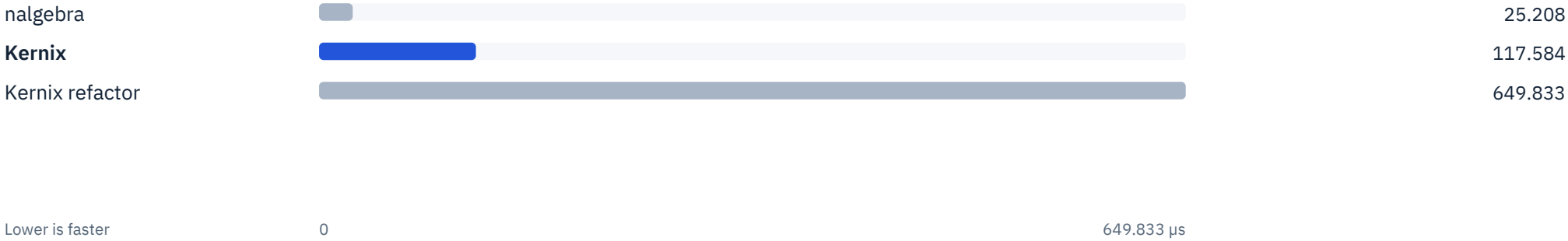
256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Cholesky rank-one downdate

Per-call median

256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



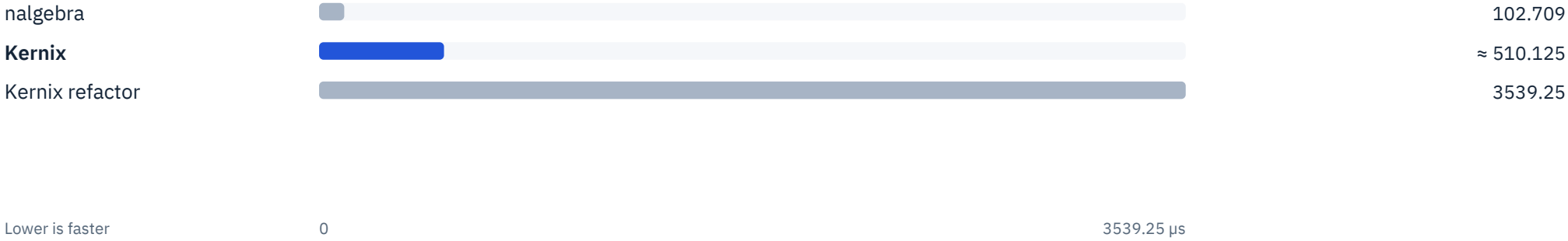
Kernix 0.13.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky rank-one update

Per-call median

512 × 512 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Thin QR

Per-call median

512 × 128 · f64 · 1 worker · allocated output

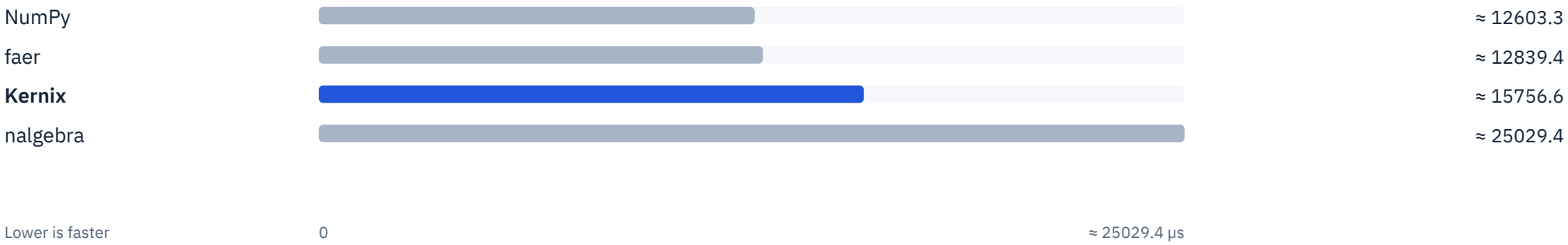


Kernix 0.13.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

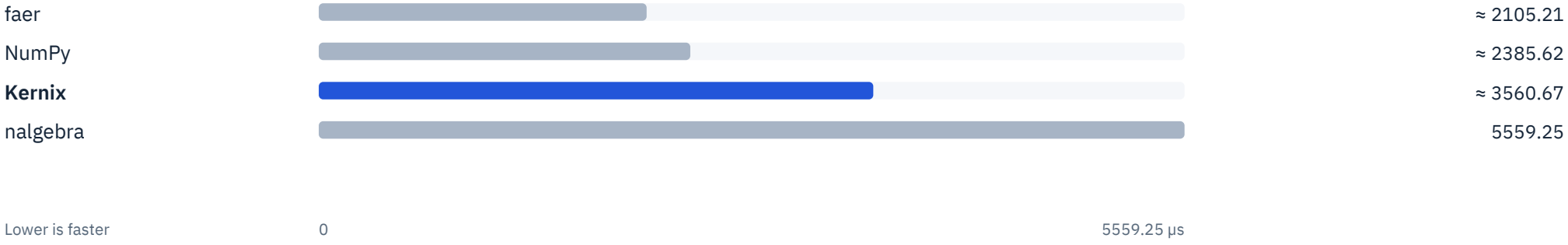
Thin QR

1024 × 256 · f64 · 1 worker · allocated output



Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output



Kernix 0.13.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

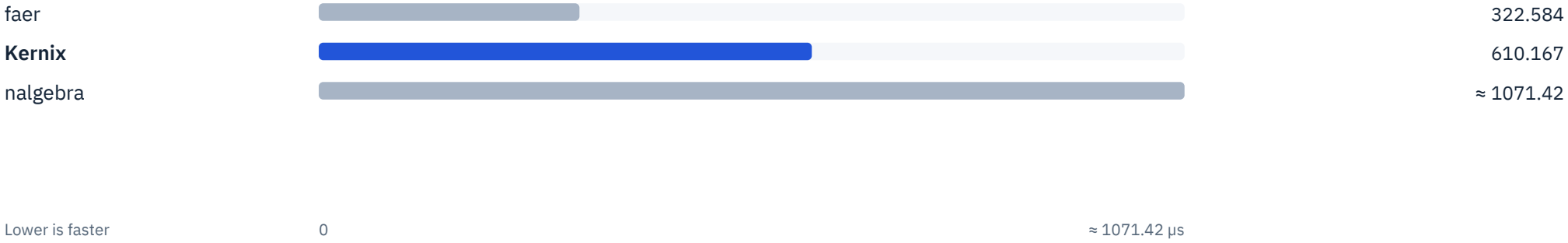
Per-call median



Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



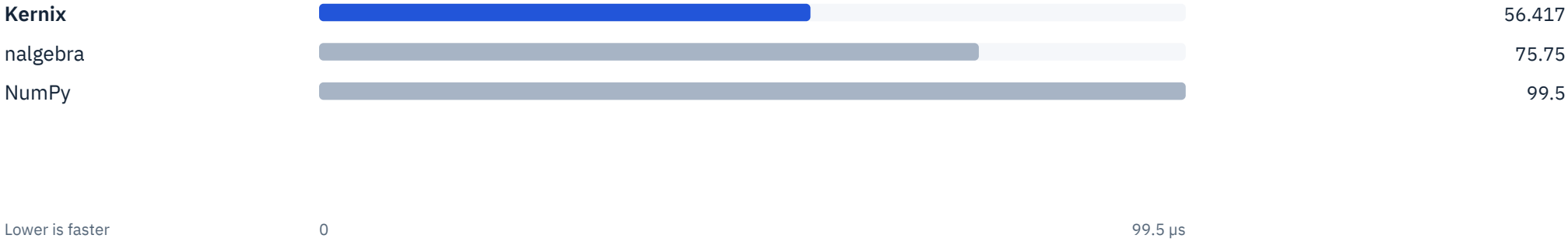
Kernix 0.13.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Minimum-norm least squares

Per-call median

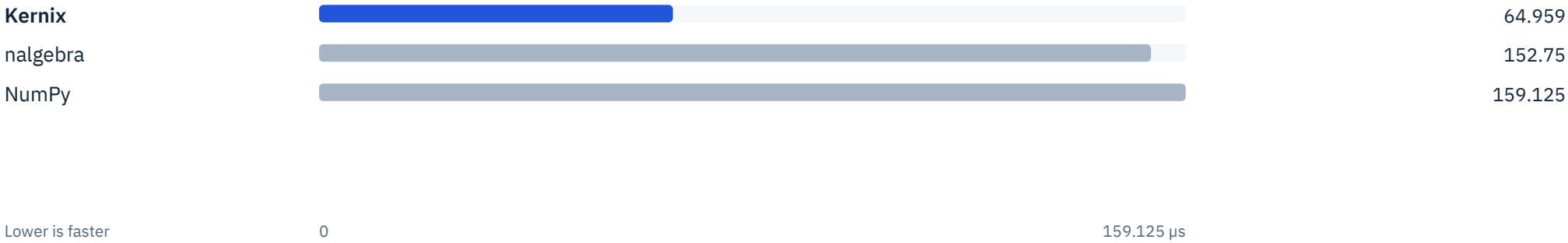
64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Kernix 0.13.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

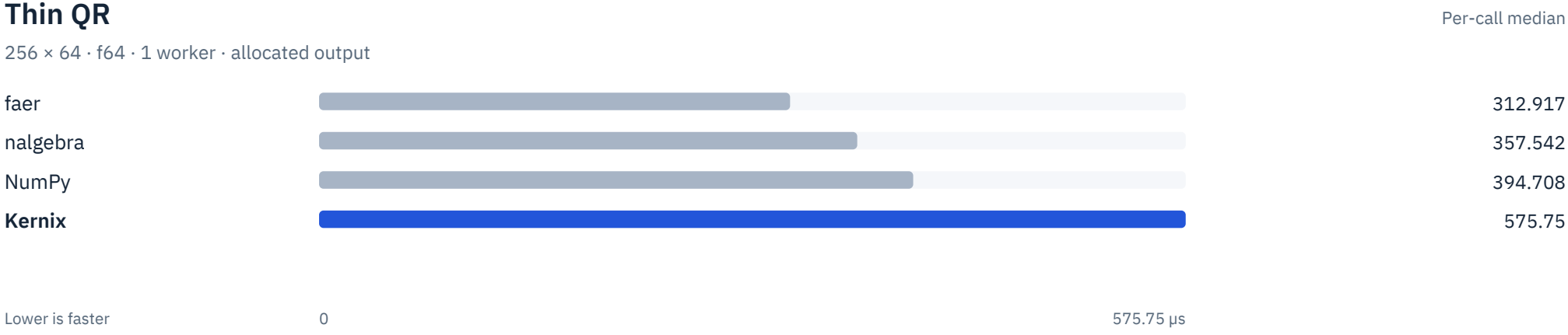
Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output



Kernix 0.13.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky factorization

Per-call median

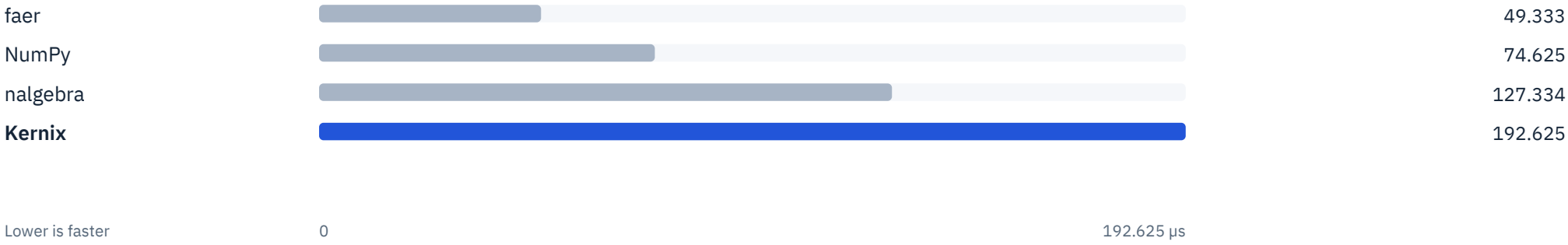
64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output

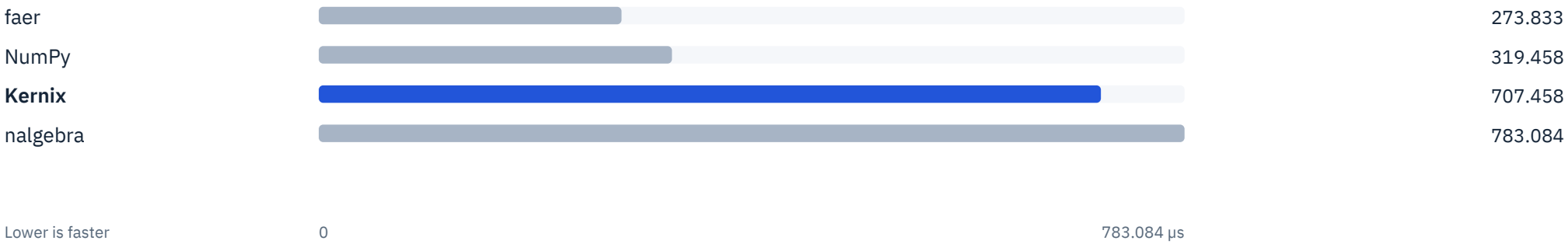


Kernix 0.13.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output



Kernix 0.13.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.13.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.13.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

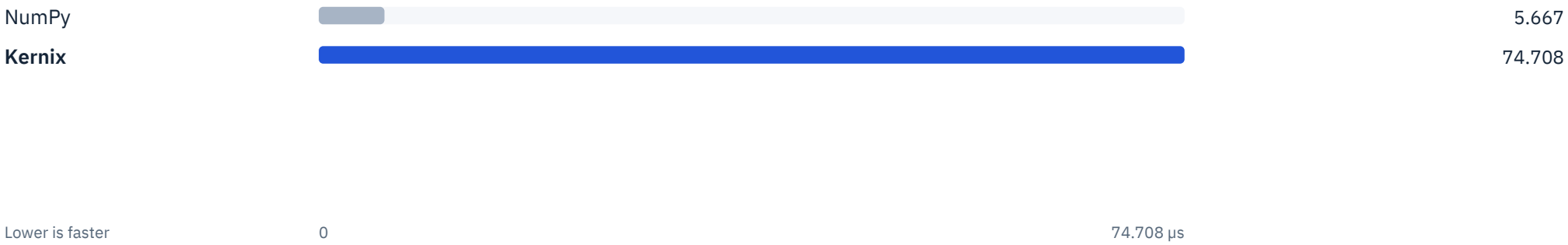


Kernix 0.13.0 / Tensor contractions

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

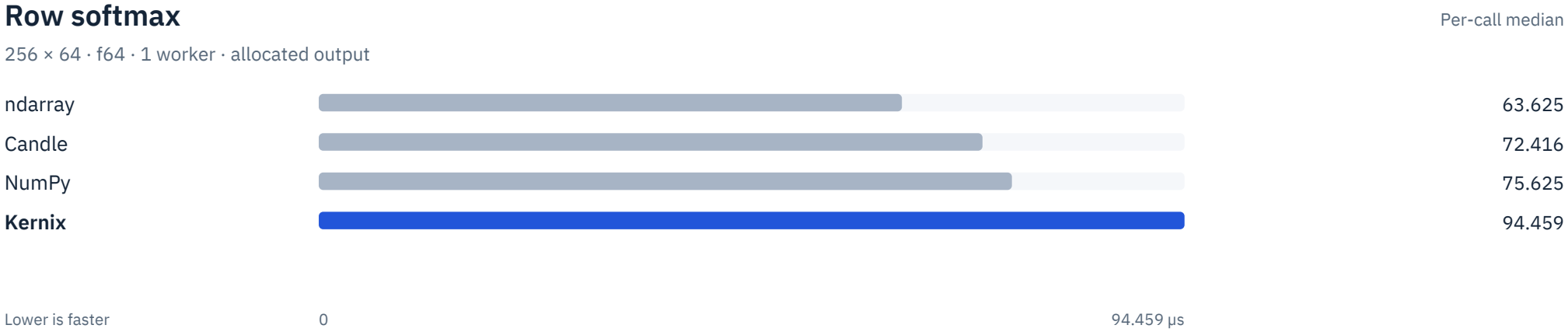


Kernix 0.13.0 / Statistics & activations

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Row softmax

256 × 64 · f64 · 1 worker · allocated output



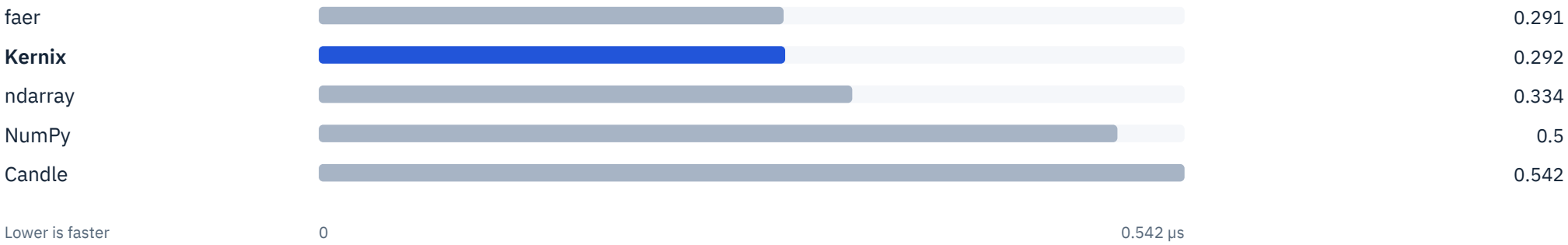
Kernix 0.13.0 / Tensor construction

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Create zeros

64 × 64 · f32 · 1 worker · allocated output

Per-call median



Create ones

64 × 64 · f32 · 1 worker · allocated output

Per-call median



Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Cholesky solve · reused factor

128 × 128, 1 RHS · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9921669382128087	0.9164682291013913 – 1.0027768595041322

Cholesky solve · reused factor

256 × 256, 64 RHS · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9885282882882883	0.8586642621355661 – 1.079443083541639

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

LU solve · reused factor

256 × 256, 64 RHS · f64

Observed time increase



CURRENT / BASELINE	95% INTERVAL
1.209485633752933	1.1546833718414344 – 1.3779067180145486

Thin QR

512 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9936245424891716	0.8976748411295378 – 1.0586904929547531

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Thin QR

1024 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0182715183366788	0.9539091491294941 – 1.0669591519324717

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0408798988621997	0.9586351199098664 – 1.0546903947744524

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9951534420254735	0.961185853240413 – 1.0564999190545572

Thin QR

256 × 64 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9986932253431998	0.9374925164989188 – 1.0351285018554286

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

32 × 32 · 32 × 32 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9821038847664775	0.945438673068529 – 1

Matrix multiplication

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.00001011807797	0.9388606238475425 – 1.0940139622787055

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9991894510013245	0.9028559547324492 – 1.0428752324799178

Matrix multiplication

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9962362276819777	0.9468969486462286 – 1.0104691648198838

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0213066491832494	0.8922478299910207 – 1.1183833501174891

Record + backward

sum(A · W), 32 × 16 · 16 × 8 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.9727626459143969	0.9375937593759376 – 0.9869077306733167

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	1	95% INTERVAL	0.8936943272386311 – 1.0455315505683473

Independent matrix operands

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	0.9991960784313726	95% INTERVAL	0.9549636566040207 – 1.0070588235294118

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9853099247581512	0.9565217391304348 – 1

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0474285714285714	0.9541984732824428 – 1.158669574700109

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9675904789281748	0.916102372139973 – 1.0378457059679767

Cholesky factorization

256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.000398406374502	0.9715976292854505 – 1.0930526315789473

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0001606038705533	0.9693697400669682 – 1.1191409331409332

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9864052621133509	0.4546756756756757 – 1.1396162361623616

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9985013499839345	0.7590126712952051 – 1.061062513641804

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9885894736842106	0.7937915742793792 – 1.2318772378516625

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Sparse matrix × vector / faer 2048 × 2048 · five diagonals · f64	6792	6.792
Sparse matrix × vector / SciPy · CSR 2048 × 2048 · five diagonals · f64	8250	8.25
Sparse matrix × vector / NumPy 2048 × 2048 · five diagonals · f64	1356042	1356.042
Sparse matrix × vector / Kernix · explicit dense 2048 × 2048 · five diagonals · f64	3004417	3004.417
Sparse matrix × vector / Kernix 2048 × 2048 · five diagonals · f64	15583	15.583
Sparse × dense matrix / SciPy · CSR 512 × 512 · 32 RHS · five diagonals · f64	21083	21.083
Sparse × dense matrix / Kernix · explicit dense 512 × 512 · 32 RHS · five diagonals · f64	440417	440.417
Sparse × dense matrix / Kernix 512 × 512 · 32 RHS · five diagonals · f64	29167	29.167
Sparse × dense matrix / NumPy 512 × 512 · 32 RHS · five diagonals · f64	87084	87.084
Sparse × dense matrix / faer 512 × 512 · 32 RHS · five diagonals · f64	54167	54.167

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Sparse × dense matrix / Kernix 512 × 512 · 32 RHS · five diagonals · f32	42583	42.583
Sparse × dense matrix / SciPy · CSR 512 × 512 · 32 RHS · five diagonals · f32	13792	13.792
Sparse × dense matrix / Kernix · explicit dense 512 × 512 · 32 RHS · five diagonals · f32	262625	262.625
Sparse × dense matrix / NumPy 512 × 512 · 32 RHS · five diagonals · f32	31458	31.458
Sparse × dense matrix / faer 512 × 512 · 32 RHS · five diagonals · f32	49584	49.584
Sparse × sparse matrix / NumPy 256 × 256 · five diagonals each · f64	129208	129.208
Sparse × sparse matrix / SciPy · CSR 256 × 256 · five diagonals each · f64	44958	44.958
Sparse × sparse matrix / Kernix · explicit dense 256 × 256 · five diagonals each · f64	827416	827.416
Sparse × sparse matrix / Kernix 256 × 256 · five diagonals each · f64	61291.99999999999	61.29199999999999
Automatic CSR construction / Kernix 512 × 512 · clone + scan + compress · f64	744500	744.5

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Automatic CSR construction / SciPy · CSR 512 × 512 · clone + scan + compress · f64	964292	964.292
Automatic CSR construction / Kernix · explicit dense 512 × 512 · clone + scan + compress · f64	53167	53.167
Automatic CSR construction / NumPy 512 × 512 · clone + scan + compress · f64	180291	180.291
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f64	14875	14.875
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f64	112625	112.625
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f32	51416	51.416
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f32	5250	5.25
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	103750	103.75
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	17459	17.459
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	5959	5.959

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	54917	54.917
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	2708	2.708
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	2250	2.25
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	2500	2.5
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	4166	4.166
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	2125	2.125
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	2041	2.041
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	14791	14.791
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	13416	13.416
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	13542	13.542

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	14084	14.084
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	14584	14.584
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	3291	3.291
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	102250	102.25
Matrix multiplication / Kernix · explicit Accelerate 128 × 128 · 128 × 128 · f64	17916	17.916
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	106583	106.583
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	17292	17.292
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	108250	108.25
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	111708	111.708
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	101417	101.417

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	124625	124.625
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	811750	811.75
Matrix multiplication / Kernix · explicit Accelerate 256 × 256 · 256 × 256 · f64	120834	120.834
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	889875	889.875
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	926958	926.958
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	811208	811.208
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	784667	784.667
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	7042	7.042
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	6917	6.917
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	5500	5.5

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	7916	7.916
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	8167	8.167
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	7250	7.25
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	51583	51.583
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	57542	57.542
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	6000	6
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	56916	56.916
Matrix multiplication / Kernix · explicit Accelerate 128 × 128 · 128 × 128 · f32	6458	6.458
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	52667	52.667
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	55291	55.291

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix · explicit Accelerate 256 × 256 · 256 × 256 · f32	35583	35.583
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	416000	416
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	427667	427.667
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	388042	388.042
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	432208	432.208
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	390541	390.541
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	32375	32.375
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	4666	4.666
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	2042	2.042
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	2833	2.833

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f32	2208	2.208
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f32	2792	2.792
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f32	1750	1.75
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	22708	22.708
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	146875	146.875
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	747042	747.042
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	55917	55.917
Cholesky solve · reused factor / Kernix 128 × 128, 1 RHS · f64	15709.000000000002	15.709000000000002
Cholesky solve · reused factor / faer 128 × 128, 1 RHS · f64	4750	4.75
Cholesky solve · reused factor / nalgebra 128 × 128, 1 RHS · f64	4708	4.708

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky solve · reused factor / faer 256 × 256, 64 RHS · f64	337000	337
Cholesky solve · reused factor / Kernix 256 × 256, 64 RHS · f64	1265292	1265.292
Cholesky solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	974459	974.459
Cholesky solve · reused factor / faer 512 × 512, 128 RHS · f64	2496000	2496
Cholesky solve · reused factor / nalgebra 512 × 512, 128 RHS · f64	7446625	7446.625
Cholesky solve · reused factor / Kernix 512 × 512, 128 RHS · f64	11043792	11043.792
LU solve · reused factor / faer 256 × 256, 64 RHS · f64	323166	323.166
LU solve · reused factor / Kernix 256 × 256, 64 RHS · f64	1430416	1430.416
LU solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	950125	950.125
Cholesky rank-one update / Kernix 256 × 256 · f64	118750	118.75

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky rank-one update / nalgebra 256 × 256 · f64	24083	24.083
Cholesky rank-one update / Kernix refactor 256 × 256 · f64	639333	639.333
Cholesky rank-one downdate / nalgebra 256 × 256 · f64	25208	25.208
Cholesky rank-one downdate / Kernix refactor 256 × 256 · f64	649833	649.833
Cholesky rank-one downdate / Kernix 256 × 256 · f64	117584	117.584
Cholesky rank-one update / Kernix refactor 512 × 512 · f64	3539250	3539.25
Cholesky rank-one update / Kernix 512 × 512 · f64	510125.00000000006	510.12500000000006
Cholesky rank-one update / nalgebra 512 × 512 · f64	102709	102.709
Thin QR / NumPy 512 × 128 · f64	2461375	2461.375
Thin QR / Kernix 512 × 128 · f64	2751375	2751.375

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / nalgebra 512 × 128 · f64	3003750	3003.75
Thin QR / faer 512 × 128 · f64	1981915.9999999998	1981.9159999999998
Thin QR / Kernix 1024 × 256 · f64	15756625	15756.625
Thin QR / nalgebra 1024 × 256 · f64	25029375	25029.375
Thin QR / faer 1024 × 256 · f64	12839417	12839.417
Thin QR / NumPy 1024 × 256 · f64	12603334	12603.334
Cholesky factorization / Kernix 512 × 512 · f64	3560667	3560.667
Cholesky factorization / NumPy 512 × 512 · f64	2385625	2385.625
Cholesky factorization / nalgebra 512 × 512 · f64	5559250	5559.25
Cholesky factorization / faer 512 × 512 · f64	2105209	2105.209

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / Kernix 96 × 32 · f64	75875	75.875
Column-pivoted QR / faer 96 × 32 · f64	43500	43.5
Column-pivoted QR / nalgebra 96 × 32 · f64	108500	108.5
Column-pivoted QR / faer 256 × 64 · f64	322584	322.584
Column-pivoted QR / nalgebra 256 × 64 · f64	1071417	1071.417
Column-pivoted QR / Kernix 256 × 64 · f64	610167	610.167
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	56417	56.417
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	99500	99.5
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	75750	75.75
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	159125	159.125

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	64959	64.959
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	152750	152.75
Thin QR / faer 96 × 32 · f64	47167	47.167
Thin QR / NumPy 96 × 32 · f64	50542	50.542
Thin QR / Kernix 96 × 32 · f64	63207.99999999999	63.20799999999999
Thin QR / nalgebra 96 × 32 · f64	41459	41.459
Thin QR / NumPy 256 × 64 · f64	394708	394.708
Thin QR / nalgebra 256 × 64 · f64	357542	357.542
Thin QR / Kernix 256 × 64 · f64	575750	575.75
Thin QR / faer 256 × 64 · f64	312917	312.917

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / faer 64 × 64 · f64	13583	13.583
Cholesky factorization / nalgebra 64 × 64 · f64	22042	22.042
Cholesky factorization / Kernix 64 × 64 · f64	21834	21.834
Cholesky factorization / NumPy 64 × 64 · f64	16042	16.042
Cholesky factorization / Kernix 128 × 128 · f64	192625	192.625
Cholesky factorization / NumPy 128 × 128 · f64	74625	74.625
Cholesky factorization / faer 128 × 128 · f64	49333	49.333
Cholesky factorization / nalgebra 128 × 128 · f64	127334	127.334
Cholesky factorization / nalgebra 256 × 256 · f64	783084	783.084
Cholesky factorization / faer 256 × 256 · f64	273833	273.833

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / NumPy 256 × 256 · f64	319458	319.458
Cholesky factorization / Kernix 256 × 256 · f64	707458	707.458
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	6000	6
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	4500	4.5
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	3792	3.792
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	6917	6.917
Model inference / Candle batched normalized/sliced MLP · f64	5792	5.792
Model inference / Kernix batched normalized/sliced MLP · f64	2250	2.25
Model recording + forward / Candle batched normalized/sliced MLP · f64	6334	6.334
Model recording + forward / Kernix batched normalized/sliced MLP · f64	3166	3.166

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model backward / Candle prepared graph → fresh gradients · f64	23209	23.209
Model backward / Kernix prepared graph → fresh gradients · f64	3375	3.375
Parameter update / Kernix transactional SGD, fixed gradients · f64	750	0.75
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	5667	5.667
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	74708	74.708
Row softmax / NumPy 256 × 64 · f64	75625	75.625
Row softmax / Candle 256 × 64 · f64	72416	72.416
Row softmax / Kernix 256 × 64 · f64	94459	94.459
Row softmax / ndarray 256 × 64 · f64	63625	63.625
Create zeros / faer 64 × 64 · f32	291	0.291

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / Candle 64 × 64 · f32	542	0.542
Create zeros / ndarray 64 × 64 · f32	334	0.334
Create zeros / Kernix 64 × 64 · f32	292	0.292
Create zeros / NumPy 64 × 64 · f32	500	0.5
Create ones / ndarray 64 × 64 · f32	375	0.375
Create ones / faer 64 × 64 · f32	416	0.416
Create ones / Candle 64 × 64 · f32	1000	1
Create ones / NumPy 64 × 64 · f32	1250	1.25
Create ones / Kernix 64 × 64 · f32	292	0.292